

Assessing Bicycle Lane Quality Using Smartphone Sensor Data

5ARE0 — Data Analysis & Learning Methods | Assignment 1 | Group 5

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1 Introduction

The CRISP-DM (CRoss-Industry Standard Process for Data Mining) framework described by [1] was used as the underlying framework for structuring the project and report. The CRISP-DM framework consists of six phases: Business Understanding (section 2), Data Understanding (section 3), Data Preparation (section 4), Modeling (section 5), Evaluation (section 6) and Deployment (section 7). The following sections of this report are structured according to these phases.

The group adopted an experimental approach to investigate the data and establish frequencies of interest related to the type of road. The empirical hypothesis was that a so-called 'bumpy' road would, by definition, present a higher energy in the band of interest due to vertical vibration, which is basically the rider's perceived shaking experience when cycling through a bumpy versus a smooth path.

Technology Statement: During the preparation of this work, we used OpenAI ChatGPT (GPT-5.6 Sol), Google Gemini (3.1 Pro) and Claude Sonnet 5 Medium in order to brainstorm assignment direction, support on coding and discuss specific technical concepts to support the team along the development. The following parts of the assignment were affected/generated by AI tool usage: DATA UNDERSTANDING / DATA PREPARATION / MODELING / EVALUATION / DEPLOYMENT. After using this tool/service, [Su Acar, Vikram Chandrasekaran and Henrique da Silva Couto] evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, [Su Acar, Vikram Chandrasekaran and Henrique da Silva Couto] take full responsibility for the content of their work.

2 Business / Objective Understanding

Research objective. The classification target was defined as *Smooth* = 0 and *Bumpy* = 1, with Bumpy treated as the positive class.

From the intended application perspective, a false positive—a Smooth road incorrectly classified as Bumpy—is considered more costly than a false negative, since repeated false alarms could unnecessarily direct inspection or maintenance resources towards roads that do not require intervention. In the intended screening workflow, a Bumpy prediction would trigger further inspection or prioritisation rather than immediate maintenance; consequently, excessive false alarms would directly consume limited inspection resources.

That being said, precision alone was proven to be not an optimal criterion. For instance, a highly conservative classifier could obtain very high precision simply by predicting *Bumpy* only in a few extreme cases, while missing most genuinely bumpy roads. Therefore, the primary model-selection metric was chosen as the $F_{0.5}$ score:

$$F_{0.5} = \frac{1.25 (\text{Precision})(\text{Recall})}{0.25 \text{Precision} + \text{Recall}}. \quad (1)$$

The value $\beta = 0.5$ weights precision more strongly than recall while still penalising models that achieve high precision by detecting too few Bumpy segments.

3 Data Understanding

3.1 Acquisition Protocol

Acquisition protocol. Smartphone sensor data were collected while participants cycled over road segments labelled as either *Smooth* or *Bumpy*. The smartphone was placed in the rider's back pocket to provide a reasonably consistent measurement position while limiting direct displacement caused by leg movement. The initial collection plan targeted at least five recordings of each terrain type per participant. Three participants originally collected data, of whom two remained after quality screening.

Three sensor streams were retained for the analysis: calibrated accelerometer, gravity and calibrated gyroscope measurements, each containing three spatial axes. Participant identities were replaced by anonymous numeric codes, while raw files followed the naming convention **RawXXX_Terrain.zip**. Raw packages were stored locally before processing rather than being retrieved automatically from external storage.

Quality screening. The effective sampling frequency of every sensor stream was estimated from the mean interval between successive timestamps. Most usable recordings were measured close to the intended 100 Hz sampling rate. Recordings originating from participant #03 were instead sampled at approximately 62 Hz.

At 62 Hz the corresponding Nyquist frequency is only approximately 31 Hz. This is incompatible with the later frequency-domain analysis (as can be seen in figure 2), which uses information extending to 46 Hz. Upsampling these recordings would increase the sample count but could not reconstruct frequency content that was never measured. Participant #03 was therefore excluded rather than mixing measurements with fundamentally different observable bandwidths. To compensate for the reduced number of participants, more data was collected as per appendix A.

After this quality-control step, 26 recordings were retained, corresponding to approximately 4789 s (79.8 min) of sensor data. This exclusion improves measurement consistency but reduces participant diversity: all subsequent modelling results are based on only two retained participants and should therefore not be interpreted as demonstrating population-level generalisation.

Available independent units. After operational-state filtering and window extraction, the retained data produced 4246 valid riding windows, comprising 2431 Smooth and 1815 Bumpy windows. These windows are not statistically independent observations because neighbouring windows overlap and several recordings originated from the same continuous acquisition session.

In particular, Raw031–Raw037 were segmented from one original annotated recording and were therefore represented by the shared group `AnnotatedSession01`. Raw038–Raw041 similarly originated from a second recording and were represented by `AnnotatedSession02`. These parent-session identifiers were subsequently used during dataset partitioning and cross-validation to prevent segments from the same original ride appearing on opposite sides of a validation split.

The measured sampling frequencies are shown in Figure 1. The clear hardware-dependent difference supports exclusion of participant #03 rather than artificial resampling, which could not recover unobserved spectral content above its Nyquist limit.

3.2 Physics-Informed Exploratory Analysis

Before defining the frequency-domain features, an exploratory experiment was performed to investigate whether bicycle-induced road vibration could be separated from low-frequency rider motion.

Road excitation was approximated from bicycle speed v and the characteristic spacing λ between surface irregularities:

$$f_{\text{road}} \approx \frac{v}{\lambda}. \quad (2)$$

For a measured irregularity spacing of $\lambda = 100$ mm, repeated 10 m runs produced estimated excitation frequencies of 25.6 Hz, 39.7 Hz and 51.2 Hz for slow, medium and fast cycling respectively. Considering the largest observed spacing ($\lambda \approx 500$ mm) together with the slowest measured speed ($v \approx 2.5$ m/s) gives an approximate lower road-excitation bound of 5 Hz.

Rider-motion frequency was independently estimated by manually counting complete pedalling cycles from video recordings:

$$f_{\text{pedal}} = \frac{N_{\text{cycles}}}{t}. \quad (3)$$

Even during intentionally fast cycling, measured pedalling frequencies ranged from approximately 1.20 to 1.53 Hz (72–92 RPM), remaining below 2 Hz. The experiment therefore indicated a useful separation between dominant pedalling motion and plausible road-induced vibration. This observation motivated treating the low-frequency region as rider-motion dominated and investigating a higher-frequency road-vibration band during subsequent signal processing.

4 Data Preparation

Processing pipeline. The three smartphone sensor streams were first aligned using their timestamps and consolidated at recording level. To reduce sensitivity to phone orientation, the analysis then transformed the original sensor axes into physically interpretable signals.

The magnitude of the gravity vector was computed and used to define the local vertical direction. Calibrated dynamic acceleration was projected onto this gravity direction to obtain `vertical_dyn_acc`, representing bicycle motion approximately normal to the road surface. The Euclidean magnitude of the three gyroscope axes was calculated as `gyroscope_mag`, while vertical jerk was obtained from the discrete time derivative of vertical dynamic acceleration.

The gyroscope magnitude was also used to distinguish stopped, transitional and active-riding periods. A smoothed gyroscope RMS envelope was compared with an empirically investigated motion threshold. Samples surrounding start/stop transitions were excluded using a 2 s transition buffer, and only windows classified as *In Movement* were retained for feature extraction. This avoids allowing stationary periods to dominate the Smooth class.

Frequency analysis and windowing. The physics experiment in Section 3.2 indicated that pedalling motion remained below approximately 2 Hz, while plausible road-excitation frequencies covered a substantially higher range. For the measured combinations of bicycle speed and terrain spacing, the estimated road excitation extended approximately from 5 to 51 Hz.

Considering both this physical argument and the approximately 100 Hz sensor sampling frequency, a fourth-order Butterworth band-pass filter from 4–46 Hz was applied to vertical dynamic acceleration. The upper limit remains below the 50 Hz Nyquist frequency of the retained recordings. The filter is used as an analysis band rather than as an assumption that every component inside 4–46 Hz is necessarily caused by road roughness.

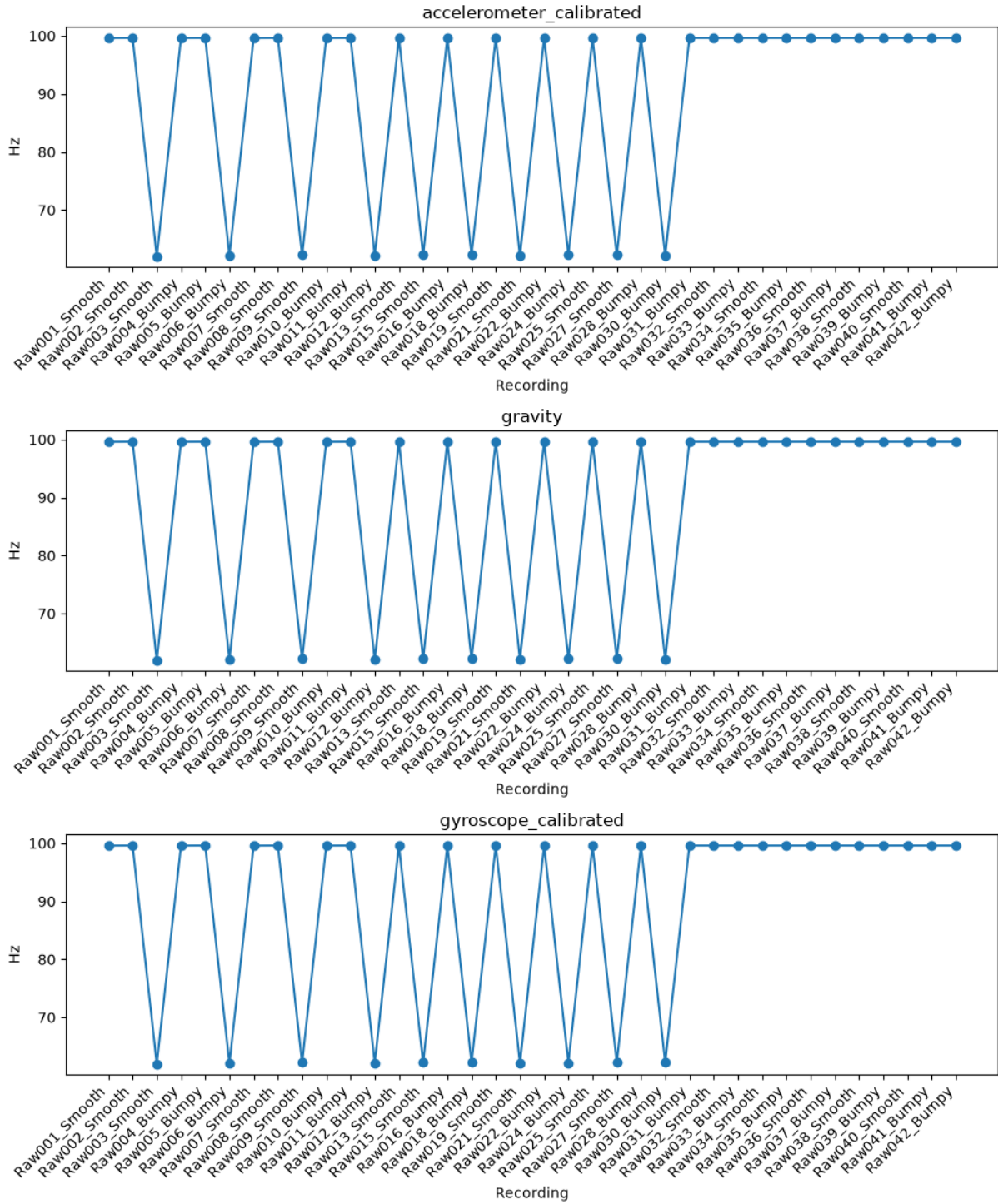


Figure 1: Measured sampling frequencies for the three recorded sensor streams across all acquired recordings. Most recordings are close to the nominal 100 Hz rate, whereas participant #03 recordings are approximately 62 Hz.

Welch power spectral density estimates were used to examine the terrain classes before modelling. As shown in Figure 2, Bumpy windows exhibit consistently greater spectral energy over much of the candidate road-vibration region. The median 4–46 Hz RMS magnitude was 1.205 m/s^2 for Bumpy windows compared with 0.673 m/s^2 for Smooth windows, corresponding to an approximately $1.79\times$ ratio.

Together with the independent pedalling experiment in Section 3.2, this supported use of the 4–46 Hz band for analysis.

Once again, the band should nevertheless be interpreted as a physically motivated analysis region rather than evidence that all energy within it is road-generated.

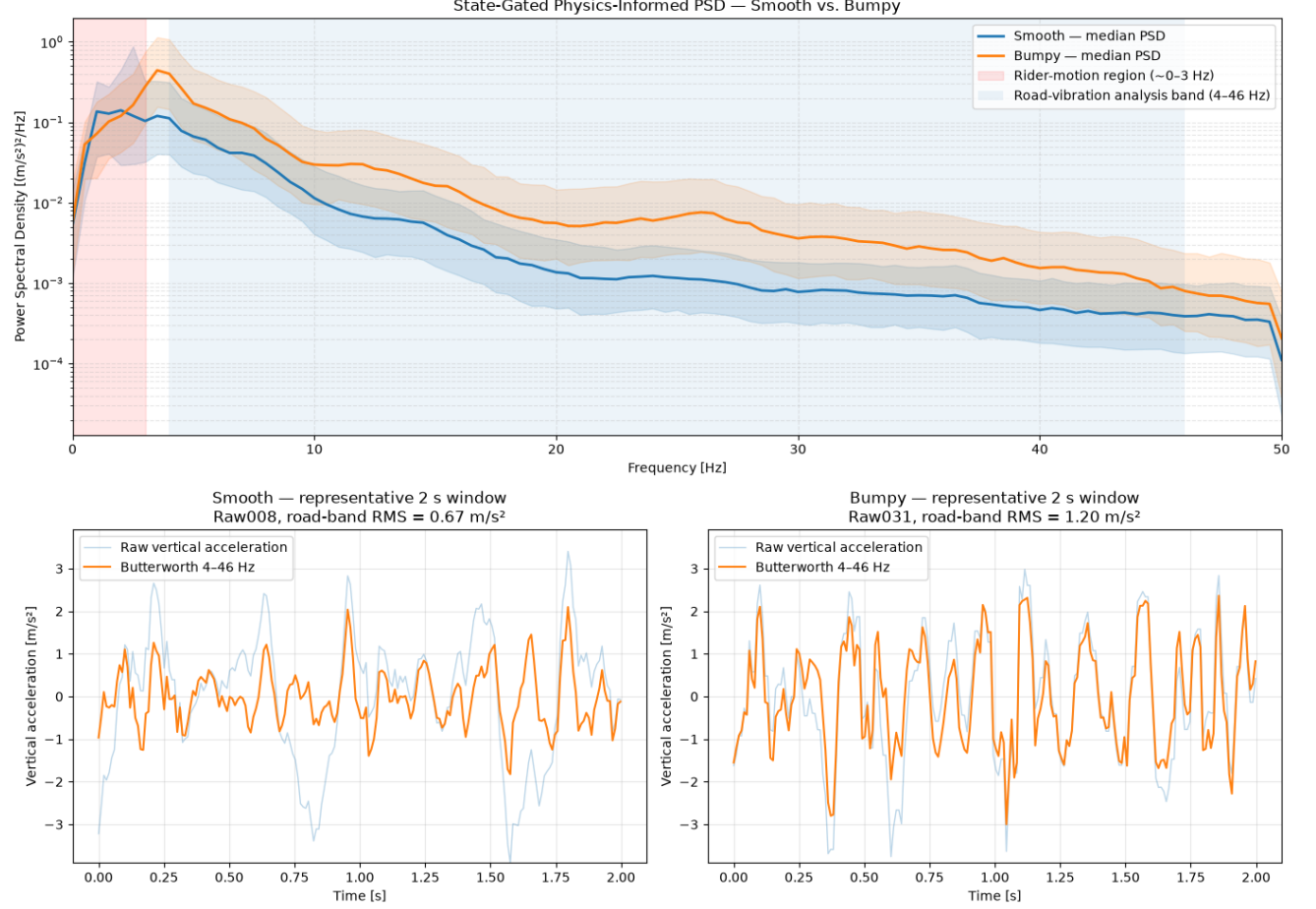


Figure 2: Physics-informed comparison of Smooth and Bumpy riding windows. The upper panel shows median Welch PSD estimates and the selected road-vibration analysis band. The lower panels illustrate representative 2 s windows before and after 4–46 Hz band-pass filtering.

The state-gated riding data (that was classified as *In Movement*) were divided into 2 s windows with 50% overlap, resulting in 4246 valid windows. Because the current filtering implementation uses non-causal processing, the present pipeline should be regarded as an offline analysis pipeline. If we consider a real-time implementation, the classifier would need to be triggered by the state-gating DSP-based algorithm. Besides, it’s worth noting that the 2 s windows with 50% overlap additionally introduce temporal dependence between neighbouring feature vectors.

Features and reduction. Twelve candidate features were engineered from each valid window: acceleration RMS, standard deviation and peak-to-peak amplitude; gyroscope RMS; vertical-jerk RMS; RMS vibration in the 4–46 Hz road band; spectral energy in the 0–3, 3–10, 10–20 and 20–40 Hz bands; total spectral energy in 4–46 Hz; and a high-to-low-frequency energy ratio.

Feature redundancy and mutual information with the terrain label were analysed using the **training partition only**. The higher-frequency and vibration-related features carried most of the information about the terrain label. The strongest mutual-information scores were obtained for 20–40 Hz spectral energy ($MI = 0.312$), jerk RMS (0.289), 4–46 Hz road-band RMS (0.252), and acceleration peak-to-peak amplitude (0.230).

Several features nevertheless represented closely related physical behaviour. Acceleration RMS and standard deviation were effectively collinear, while jerk RMS and road-band RMS showed $|r| = 0.954$. Because several of the strongest features were also highly correlated, we did not simply keep the features with the highest individual scores. Instead, we compared three feature sets during modelling.

Three candidate representations were evaluated: **full_12**, containing all twelve features; **compact_physics_5**, containing five physically motivated features; and **compact_ratio_4**, containing four features including the high/low spectral-energy ratio.

Interestingly, Logistic Regression still performed best with all 12 features, despite the redundancy between some of them. k-NN behaved differently and performed best with the reduced `compact_physics_5` set (see figure 10).

5 Modeling

Partitioning. A deterministic recording-level hold-out was frozen before model fitting. Complete recordings Raw001 and Raw022 from participant #01 and Raw008 and Raw011 from participant #02 were assigned to the final test set. Each participant therefore contributed both a Smooth and a Bumpy independent recording to the hold-out.

The resulting partition contained 3311 training windows (78.0%) and 935 held-out test windows (22.0%). The training partition contained 2007 Smooth and 1304 Bumpy windows, while the final test partition contained 424 Smooth and 511 Bumpy windows.

All windows associated with an original acquisition session were represented by the same grouping variable. Consequently, the training set contained 13 independent split groups rather than treating all 3311 windows as independent samples.

Model development used five-fold `StratifiedGroupKFold` cross-validation on the training set. This simultaneously attempted to preserve class representation and prevented windows from the same recording or parent session from appearing in both the training and validation portions of a fold. Scaling was implemented inside each model pipeline so that transformation parameters were learned from the corresponding training fold only.

Candidate feature sets. Three predefined feature representations were compared for every model: `full_12`, containing all twelve engineered features; `compact_physics_5`, containing gyroscope RMS, 4–46 Hz road-band RMS and spectral energies at 0–3, 10–20 and 20–40 Hz; and `compact_ratio_4`, containing gyroscope RMS, road-band RMS, 20–40 Hz energy and the high/low-frequency energy ratio.

The comparison was designed to determine whether compact, physically interpretable feature sets could match or improve the performance of the complete representation, particularly given the strong correlations observed between several candidate features.

Supervised methods. Four supervised methods were compared, each representing a different type of classifier. k-nearest neighbours (k-NN) tested whether Smooth and Bumpy observations form different local neighbourhoods in feature space. Logistic Regression provided a regularised linear baseline. A Decision Tree allowed nonlinear threshold rules to be learned directly, while Random Forest tested whether aggregating multiple trees could capture more robust feature interactions.

For k-NN, k was searched from 3 to 49 in steps of two, with uniform and distance weighting and Manhattan ($p = 1$) or Euclidean ($p = 2$) distance. For Logistic Regression, $C \in \{0.01, 0.1, 1, 10\}$. Candidate preprocessing for both methods included no scaling, `StandardScaler`, `RobustScaler`, `MinMaxScaler` and `PowerTransformer`. Decision Tree depth was searched over $\{2, 3, 4, 5, 6, 8, 10, 12, 15, \text{None}\}$ with minimum leaf sizes from 1 to 100. Random Forest used 100 or 200 trees, maximum depths from 3 through 20 or unlimited depth, and minimum leaf sizes from 1 to 50.

Hyperparameter selection used grouped training-only cross-validation with $F_{0.5}$ as the refit criterion. After hyperparameter selection, out-of-fold training probabilities were additionally used to select the classification threshold.

Unsupervised methods. Four clustering approaches were also investigated without using terrain labels as fitting inputs. The four clustering methods were chosen because they look for structure in different ways: K-Means is centroid-based, Agglomerative Clustering is hierarchical, DBSCAN is density-based, and GMM models overlapping probabilistic components. The number of clusters/components was investigated between 2 and 6 where applicable. DBSCAN searched ϵ between 0.3 and 2.0 and minimum sample counts between 5 and 50, while Agglomerative Clustering compared Ward, complete and average linkage. Four scaling transformations were considered for the unsupervised methods.

The best configuration within each clustering family was selected using Silhouette score only. Ground-truth terrain labels were introduced only *after fitting* to calculate ARI, NMI and mapped classification diagnostics; they therefore did not determine the clusters themselves.

Table 1: Best training-validation configuration from each model family.

Approach	Method	Selected configuration / evidence
Supervised	k-NN	<code>compact_physics_5</code> ; <code>PowerTransformer</code> ; $k = 47$; Manhattan distance; threshold 0.83; OOF $F_{0.5} = 0.848$
Supervised	Logistic Regression	<code>full_12</code> ; <code>RobustScaler</code> ; $C = 0.1$; threshold 0.68; OOF $F_{0.5} = 0.855$
Supervised	Decision Tree	<code>full_12</code> ; selected depth 4 configuration; threshold 0.90; OOF $F_{0.5} = 0.828$
Supervised	Random Forest	<code>full_12</code> ; selected depth 3 configuration; threshold 0.74; OOF $F_{0.5} = 0.847$
Unsupervised	K-Means	<code>compact_physics_5</code> ; 2 clusters; Silhouette = 0.777, ARI = 0.045, NMI = 0.067
Unsupervised	Agglomerative	<code>compact_ratio_4</code> ; average linkage, 2 clusters; Silhouette = 0.927, ARI = 0.001, NMI = 0.001
Unsupervised	DBSCAN	<code>compact_ratio_4</code> ; $\epsilon = 1.0$, minimum samples = 10; noise fraction = 0.070, Silhouette = 0.750
Unsupervised	GMM	<code>compact_physics_5</code> ; tied covariance, 2 components; Silhouette = 0.855, ARI = 0.010, NMI = 0.015

6 Evaluation

Selection criterion. Bumpy was consistently treated as the positive class. Model development was conducted exclusively on the training partition, with $F_{0.5}$ used as the primary selection criterion because false Bumpy alarms are considered more costly than missed detections, while non-zero recall remains necessary for a useful classifier.

All five grouped validation folds contained examples of both classes, so class-dependent metrics including ROC - AUC were mathematically defined for each fold. Precision, recall, specificity, F_1 , balanced accuracy, ROC - AUC and average precision were nevertheless interpreted as supporting diagnostics rather than independently optimised objectives.

Supervised comparison. Each of the four supervised families was evaluated using all three candidate feature representations under the same frozen grouped cross-validation scheme. Hyperparameters were fitted inside the grouped folds, after which training-only out-of-fold probabilities were used to determine a decision threshold for each candidate.

The strongest configuration from each model family achieved training out-of-fold $F_{0.5}$ values of 0.855 for Logistic Regression, 0.848 for k-NN, 0.847 for Random Forest and 0.828 for Decision Tree. The differences between the first three approaches were therefore relatively small, but Logistic Regression provided the highest validation criterion.

The resulting final specification was consequently frozen as Logistic Regression using the complete `full_12` feature set, RobustScaler, $C = 0.1$, and a Bumpy probability threshold of 0.68. Both the model specification and the threshold were fixed before accessing the held-out test recordings.

Final model rationale. Logistic Regression was selected not only because it achieved the highest training-only OOF $F_{0.5}$, but also because its characteristics are well matched to the intended application. The problem uses a compact engineered feature representation rather than raw high-dimensional signals, making a regularised linear classifier appropriate for the available dataset size. Its coefficients and probability outputs are also more directly interpretable than those of the ensemble alternative, while inference requires only a small number of arithmetic operations per window. This makes the model comparatively lightweight for future mobile or edge deployment. Random Forest and k-NN were very close in validation performance, so there was little reason to choose the more complex alternative over Logistic Regression. The latter is also easier to interpret and lightweight enough for a future mobile or edge implementation.

For the selected Logistic Regression model, `full_12` was ultimately the optimal feature representation under grouped training-only $F_{0.5}$ validation, despite the observed redundancy; the strongest individual training-only MI features were 20–40 Hz energy, jerk RMS, 4–46 Hz road-band RMS and acceleration peak-to-peak.

Clustering comparison. The unsupervised analysis produced a different result depending on whether internal cluster compactness or agreement with the known terrain classes was considered. Agglomerative Clustering obtained the highest Silhouette score (0.927), but its ARI and NMI were only approximately 0.001, indicating that its very well-separated clusters did not correspond meaningfully to the Smooth/Bumpy distinction.

Similarly, GMM achieved a Silhouette score of 0.855 but only $\text{ARI} = 0.010$ and $\text{NMI} = 0.015$. K-Means showed somewhat greater correspondence with the terrain labels ($\text{ARI} = 0.045$, $\text{NMI} = 0.067$), while DBSCAN produced the highest diagnostic agreement among the clustering approaches ($\text{ARI} = 0.060$, $\text{NMI} = 0.069$) and classified approximately 7% of the training observations as noise.

The high Silhouette scores show that there is clear structure in the engineered feature space, but the very low ARI and NMI values show that this structure does not correspond well to Smooth versus Bumpy terrain. In other words, a high Silhouette score does not necessarily mean that the clusters represent the terrain classes we care about. Furthermore, Agglomerative Clustering does not naturally provide an out-of-sample prediction rule, while standard DBSCAN cannot directly assign arbitrary future observations without an additional deployment mechanism.

Frozen hold-out evaluation. After all model, feature and threshold decisions had been frozen, the four independent held-out recordings were evaluated once.

The final Logistic Regression classifier obtained accuracy = 0.565, Bumpy precision = 0.964, recall = 0.211, specificity = 0.991, $F_{0.5} = 0.563$, $F_1 = 0.347$, balanced accuracy = 0.601, ROC - AUC = 0.947, and average precision = 0.942.

Figure 3 makes the selected operating trade-off explicit: only 4 of 424 Smooth windows were falsely flagged as Bumpy, while 403 of 511 Bumpy windows were missed. The classifier therefore achieved the intended low false-positive behaviour, but at the cost of substantial false negatives.

The reduction from OOF $F_{0.5} = 0.855$ to held-out $F_{0.5} = 0.563$ also shows that grouped training validation underestimated the difficulty of generalising to some unseen recordings.

This weakness was not uniform across the hold-out set. Raw001 classified all 46 Smooth windows correctly, while Raw008 achieved 98.9% accuracy. The two Bumpy recordings behaved very differently: Raw022 achieved 87.5% accuracy with mean Bumpy probability 0.868, whereas Raw011 achieved only 11.6% accuracy with mean probability 0.414.

This suggests recording- or condition-dependent distribution shift rather than a global inability to rank rough terrain. The high ROC - AUC together with low thresholded recall also indicates that the fixed 0.68 threshold was conservative on the held-out distribution. It should not, however, be retrospectively tuned using this test set.

Supervised versus unsupervised behaviour. Supervised learning explicitly tries to separate the known Smooth and Bumpy classes, whereas clustering searches for structure without knowing those labels. Supervised learning directly optimises separation of the predefined Smooth and Bumpy classes, whereas clustering searches for structure in feature space without assuming that terrain condition is the dominant source of variation.

For this task, supervised learning is therefore preferable because the target classes are explicitly known and deployment

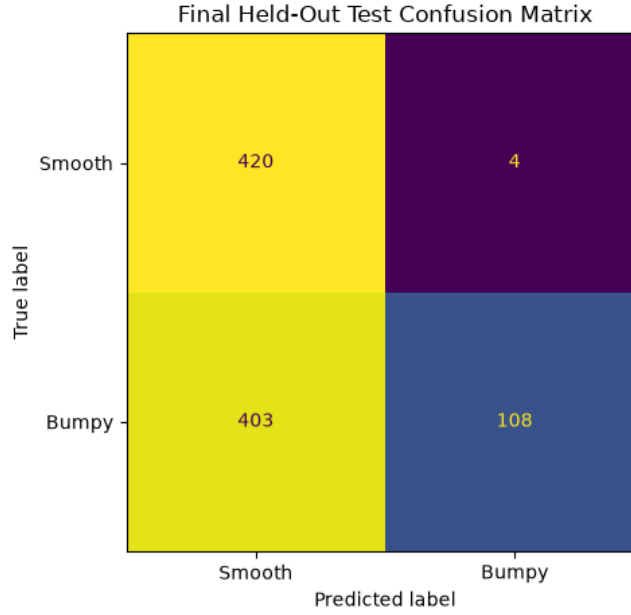


Figure 3: Confusion matrix for the final frozen Logistic Regression model evaluated once on the four held-out recordings.

requires direct classification. The unsupervised models remain useful as exploratory tools for revealing latent structure, identifying atypical observations and testing whether the known terrain labels correspond to natural groupings. The weak correspondence between clusters and terrain labels suggests that other factors - such as rider, recording session, cycling speed, phone orientation or vibration intensity - also contribute substantially to the structure of the feature space.

7 Deployment Considerations & Recommendations

External deployment. Following the one-time held-out evaluation, the final pipeline was applied without modification to the independent external dataset. The external data were not used for training, hyperparameter selection, threshold selection or testing.

The three required sensor streams were sampled at approximately 99.48 Hz, consistent with the retained training recordings. After applying the same signal derivation, operational-state gating, 4–46 Hz filtering, 2 s windowing and 12-feature extraction procedure, 888 valid riding windows remained.

The frozen RobustScaler and Logistic Regression model were then applied with the previously selected threshold of 0.68. Figure 5 shows that most windows were predicted Smooth, interspersed with short intervals where Bumpy probability exceeded the decision threshold.

Because the external recording contains no terrain ground truth, these results cannot be used to calculate classification performance. They instead demonstrate that the complete preprocessing and inference pipeline can be transferred to unseen sensor data and produce Smooth/Bumpy predictions over time, which would be suitable for a later stage real-time smartphone deployment.

7.1 Limitations and Recommendations

The main limitation is the small number of independent acquisition units. Although 4246 valid windows were extracted, overlapping windows from the same ride are correlated and should not be interpreted as independent experiments. After participant #03 was excluded because of sampling-rate incompatibility, only two participants remained.

The held-out results also indicate substantial recording-dependent generalisation. In particular, the two unseen Bumpy recordings produced very different outcomes: Raw022 achieved 87.5% accuracy with mean Bumpy probability 0.868, whereas Raw011 achieved only 11.6% accuracy with mean probability 0.414. Future data collection should therefore prioritise new participants, devices, routes, riding speeds and acquisition sessions rather than additional overlapping windows from existing rides.

A deployable implementation should also standardise a minimum sensor sampling rate or redesign the spectral representation for devices unable to observe the current 4–46 Hz band. Finally, the present preprocessing contains offline/non-causal operations. Real-time smartphone deployment would require causal filtering and explicit evaluation of latency, computational cost and any resulting change in classification performance.

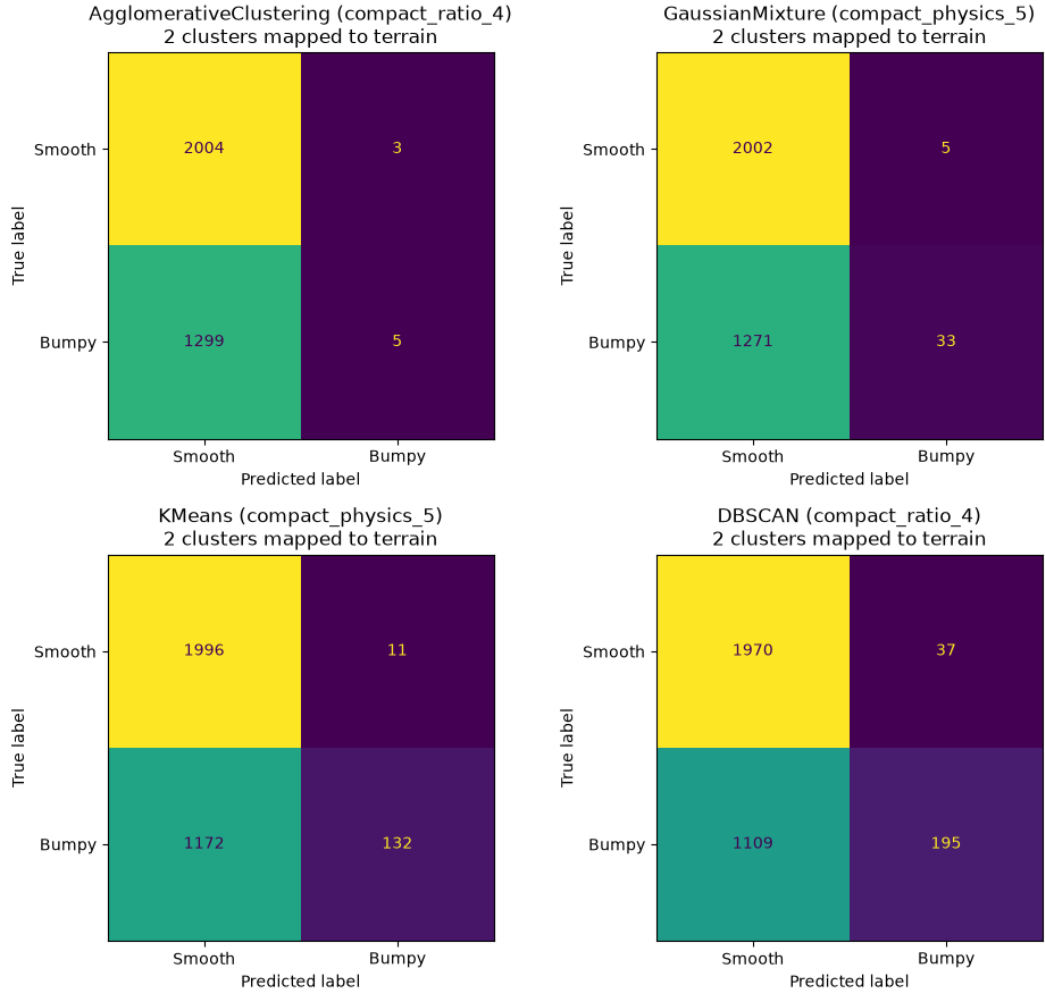


Figure 4: Mapped training-set confusion matrices for the best configuration from each unsupervised clustering family. Terrain labels were not used during clustering and were applied only afterwards to map clusters for diagnostic comparison.

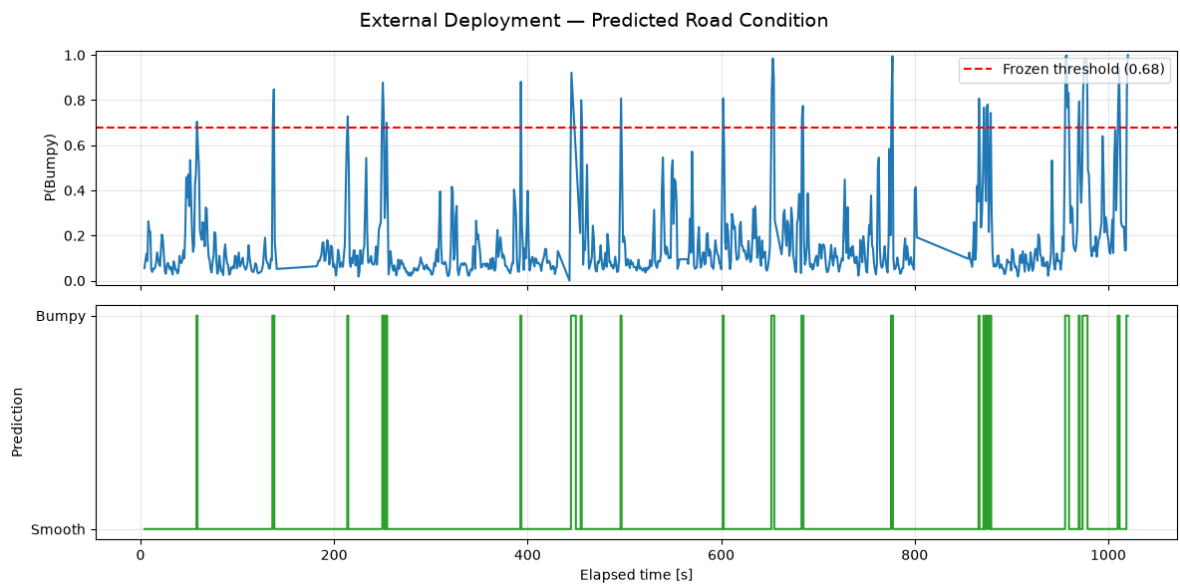


Figure 5: Deployment of the frozen Logistic Regression model on the independent external recording. The upper panel shows predicted Bumpy probability and the fixed threshold of 0.68; the lower panel shows the corresponding Smooth/Bumpy classification over elapsed time.

References

- [1] Colin Shearer. “The CRISP-DM Model: The New Blueprint for Data Mining”. In: *Journal of Data Warehousing* 5.4 (2000), pp. 13–22.

Appendices

A Data Collection Strategy

Placement. The smartphone was placed in the participant’s back pocket. This placement was chosen to reduce direct leg displacement effects and to maintain a consistent sensor position across participants.

Initial collection plan. The initial target was to collect five recordings of each road condition per participant, resulting in a planned total of 30 datasets. Each recording was intended to have a duration of at least 30 s.

Data anonymisation. Participants were assigned secret codes that were not associated with their names in the final assignment material. Raw recordings followed the naming convention

`<Sample_ID>_<path_condition>.zip`

For example:

`Raw001_Smooth.zip`

Participant names are not shown in order to preserve anonymisation.

Table 2: Raw data collection overview.

Sample ID	Participant code	Path condition	Uploaded file
Raw001	#01	Smooth	Raw001_Smooth.zip
Raw002	#02	Smooth	Raw002_Smooth.zip
Raw003	#03	Smooth	Raw003_Smooth.zip
Raw004	#01	Bumpy	Raw004_Bumpy.zip
Raw005	#02	Bumpy	Raw005_Bumpy.zip
Raw006	#03	Bumpy	Raw006_Bumpy.zip
Raw007	#01	Smooth	Raw007_Smooth.zip
Raw008	#02	Smooth	Raw008_Smooth.zip
Raw009	#03	Smooth	Raw009_Smooth.zip
Raw010	#01	Bumpy	Raw010_Bumpy.zip
Raw011	#02	Bumpy	Raw011_Bumpy.zip
Raw012	#03	Bumpy	Raw012_Bumpy.zip
Raw013	#01	Smooth	Raw013_Smooth.zip
Raw014	#02	Smooth	not collected
Raw015	#03	Smooth	Raw015_Smooth.zip
Raw016	#01	Bumpy	Raw016_Bumpy.zip
Raw017	#02	Bumpy	not collected
Raw018	#03	Bumpy	Raw018_Bumpy.zip
Raw019	#01	Smooth	Raw019_Smooth.zip
Raw020	#02	Smooth	not collected
Raw021	#03	Smooth	Raw021_Smooth.zip
Raw022	#01	Bumpy	Raw022_Bumpy.zip
Raw023	#02	Bumpy	not collected
Raw024	#03	Bumpy	Raw024_Bumpy.zip
Raw025	#01	Smooth	Raw025_Smooth.zip
Raw026	#02	Smooth	not collected
Raw027	#03	Smooth	Raw027_Smooth.zip
Raw028	#01	Bumpy	Raw028_Bumpy.zip
Raw029	#02	Bumpy	not collected
Raw030	#03	Bumpy	Raw030_Bumpy.zip
Raw031	#01	Bumpy	Raw031_Bumpy.zip
Raw032	#01	Smooth	Raw032_Smooth.zip
Raw033	#01	Bumpy	Raw033_Bumpy.zip
Raw034	#01	Smooth	Raw034_Smooth.zip
Raw035	#01	Bumpy	Raw035_Bumpy.zip
Raw036	#01	Smooth	Raw036_Smooth.zip
Raw037	#01	Bumpy	Raw037_Bumpy.zip
Raw038	#01	Smooth	Raw038_Smooth.zip
Raw039	#01	Bumpy	Raw039_Bumpy.zip
Raw040	#01	Smooth	Raw040_Smooth.zip
Raw041	#01	Bumpy	Raw041_Bumpy.zip
Raw042	#01	Bumpy	Raw042_Bumpy.zip

A.1 Additional Annotated Recordings

On 20 September 2026, additional data were collected by participant #01 (pt01) to compensate for missing recordings and the acquisition issue observed with participant #03. Two longer annotated recordings were added and subsequently segmented into individual Smooth and Bumpy samples.

Because multiple segmented files originated from the same continuous source recording, they were kept under a shared split-group identifier. This grouping was later used during model partitioning and cross-validation to prevent segments from the same original ride from appearing in both training and validation subsets.

Table 3: Segmented recordings derived from the additional annotated data.

Recording ID	Terrain	Source / split group	Output file
Raw031	Bumpy	pt01-annotations-19min-2026-09-19_09-02-09.zip	Raw031_Bumpy.zip
Raw032	Smooth	pt01-annotations-19min-2026-09-19_09-02-09.zip	Raw032_Smooth.zip
Raw033	Bumpy	pt01-annotations-19min-2026-09-19_09-02-09.zip	Raw033_Bumpy.zip
Raw034	Smooth	pt01-annotations-19min-2026-09-19_09-02-09.zip	Raw034_Smooth.zip
Raw035	Bumpy	pt01-annotations-19min-2026-09-19_09-02-09.zip	Raw035_Bumpy.zip
Raw036	Smooth	pt01-annotations-19min-2026-09-19_09-02-09.zip	Raw036_Smooth.zip
Raw037	Bumpy	pt01-annotations-19min-2026-09-19_09-02-09.zip	Raw037_Bumpy.zip
Raw038	Smooth	pt01-annotations-19min-2026-09-19_10-44-41.zip	Raw038_Smooth.zip
Raw039	Bumpy	pt01-annotations-19min-2026-09-19_10-44-41.zip	Raw039_Bumpy.zip
Raw040	Smooth	pt01-annotations-19min-2026-09-19_10-44-41.zip	Raw040_Smooth.zip
Raw041	Bumpy	pt01-annotations-19min-2026-09-19_10-44-41.zip	Raw041_Bumpy.zip

B Empirical Measurements - Road Frequency versus Leg Movement Artifacts

This appendix provides the raw experimental measurements supporting the physics-informed frequency analysis summarized in Section 3.2.



Figure 6: Example of manually counting the rotations per minute based on the video recording from one of the participants.

Table 4: Measured cycling speed and estimated road-excitation frequency for a terrain spacing of $\lambda = 100$ mm.

λ (mm)	Distance (m)	Intended speed	t_1 (s)	t_2 (s)	t_3 (s)	$\bar{t}$ (s)	Speed (m/s)	f_{road} (Hz)
100	10	Slow	3.63	4.06	4.03	3.91	2.56	25.60
100	10	Medium	2.50	2.45	2.60	2.52	3.97	39.74
100	10	Fast	1.95	1.85	2.06	1.95	5.12	51.19

C Data Acquisition and Signal-Processing Diagnostics

This appendix presents additional diagnostic evidence supporting the data-quality screening, operational-state detection and physics-informed signal-processing decisions described in Sections 3 and 4.

D Model Validation Diagnostics

This appendix provides the detailed confusion matrices corresponding to the training-only supervised and unsupervised model comparisons discussed in Section 6.



Figure 7: Different types of bumpy terrains were considered. The measured terrains ranged from 100 ~ 500mm gap interval.

Table 5: Estimated pedalling cadence from manually counted leg-motion cycles.

Video	Duration (s)	Cycles counted	f_{pedal} (Hz)	Cadence (RPM)
rpm-video-1.mp4	17	26	1.53	91.76
rpm-video-2.mp4	21	30	1.43	85.71
rpm-video-3.mp4	25	30	1.20	72.00

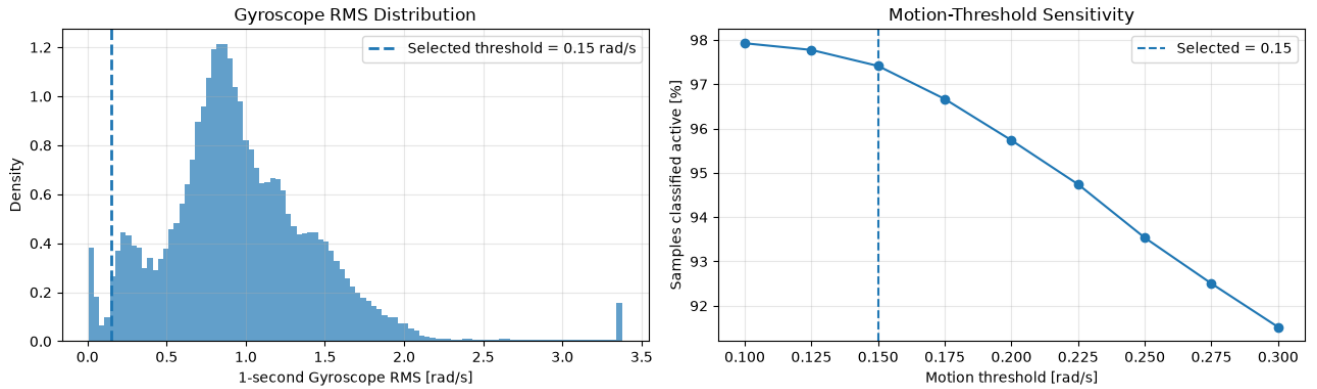


Figure 8: Diagnostic analysis used to select the gyroscope-based motion threshold. The left panel shows the distribution of the one-second gyroscope RMS envelope and the selected threshold of 0.15 rad/s. The right panel shows the sensitivity of the proportion of samples classified as active to alternative threshold values.

D.1 Supervised out-of-fold comparison

Across the supervised candidates, all selected operating points favour a low false-positive rate for Bumpy predictions. Logistic Regression provided the best $F_{0.5}$ compromise, with 66 Smooth windows incorrectly classified as Bumpy and 848 correctly detected Bumpy windows in the combined out-of-fold training predictions.

D.2 Unsupervised comparison

The mapped confusion matrices reinforce the distinction between internal cluster separation and terrain-class agreement. Agglomerative Clustering and Gaussian Mixture Modelling formed internally well-separated structures, but these structures mapped predominantly to the Smooth class. DBSCAN produced the greatest correspondence with the Smooth/Bumpy target among the evaluated clustering methods, although its agreement remained weak compared with the supervised approaches. See figure 4

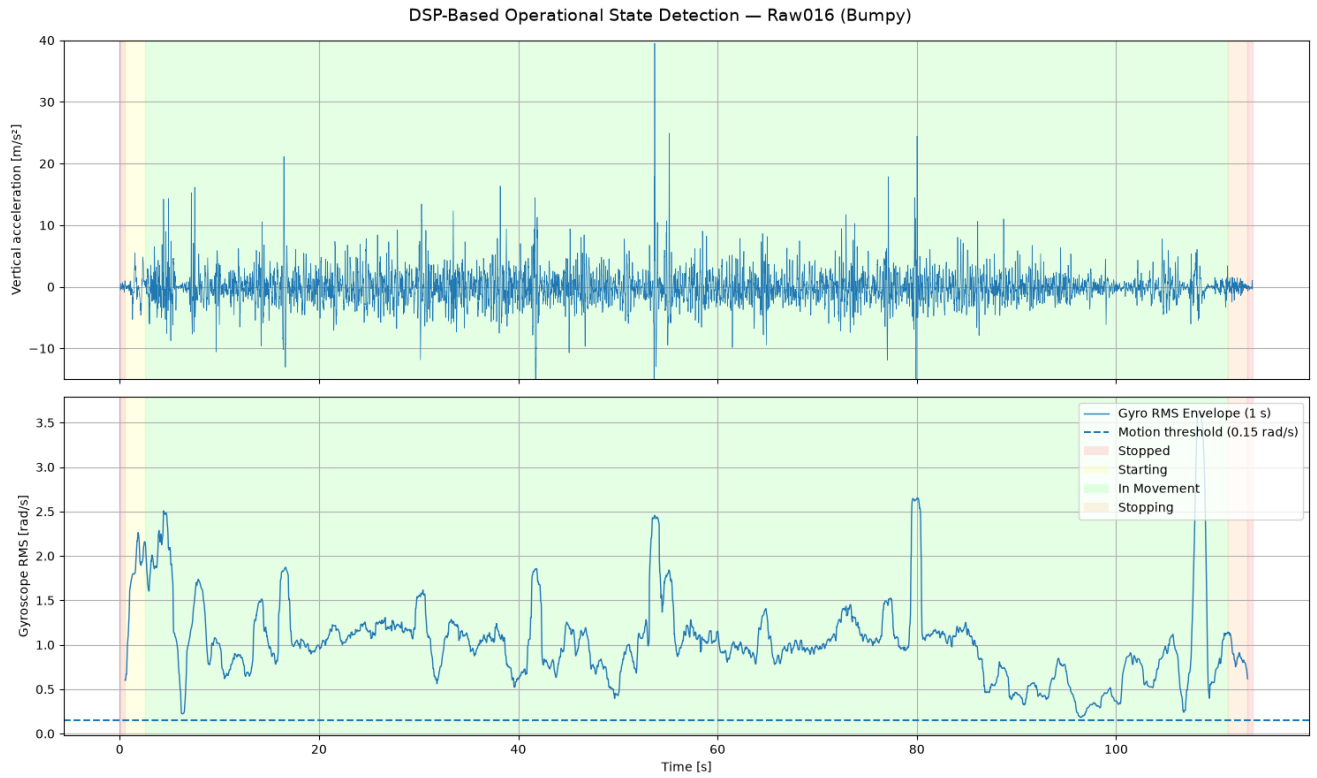


Figure 9: Example of DSP-based operational-state detection for a Bumpy recording. The upper panel shows vertical dynamic acceleration and the lower panel the smoothed gyroscope RMS envelope. Background shading marks samples classified as Stopped, Starting, In Movement and Stopping. Only stable In Movement regions were retained for feature extraction.

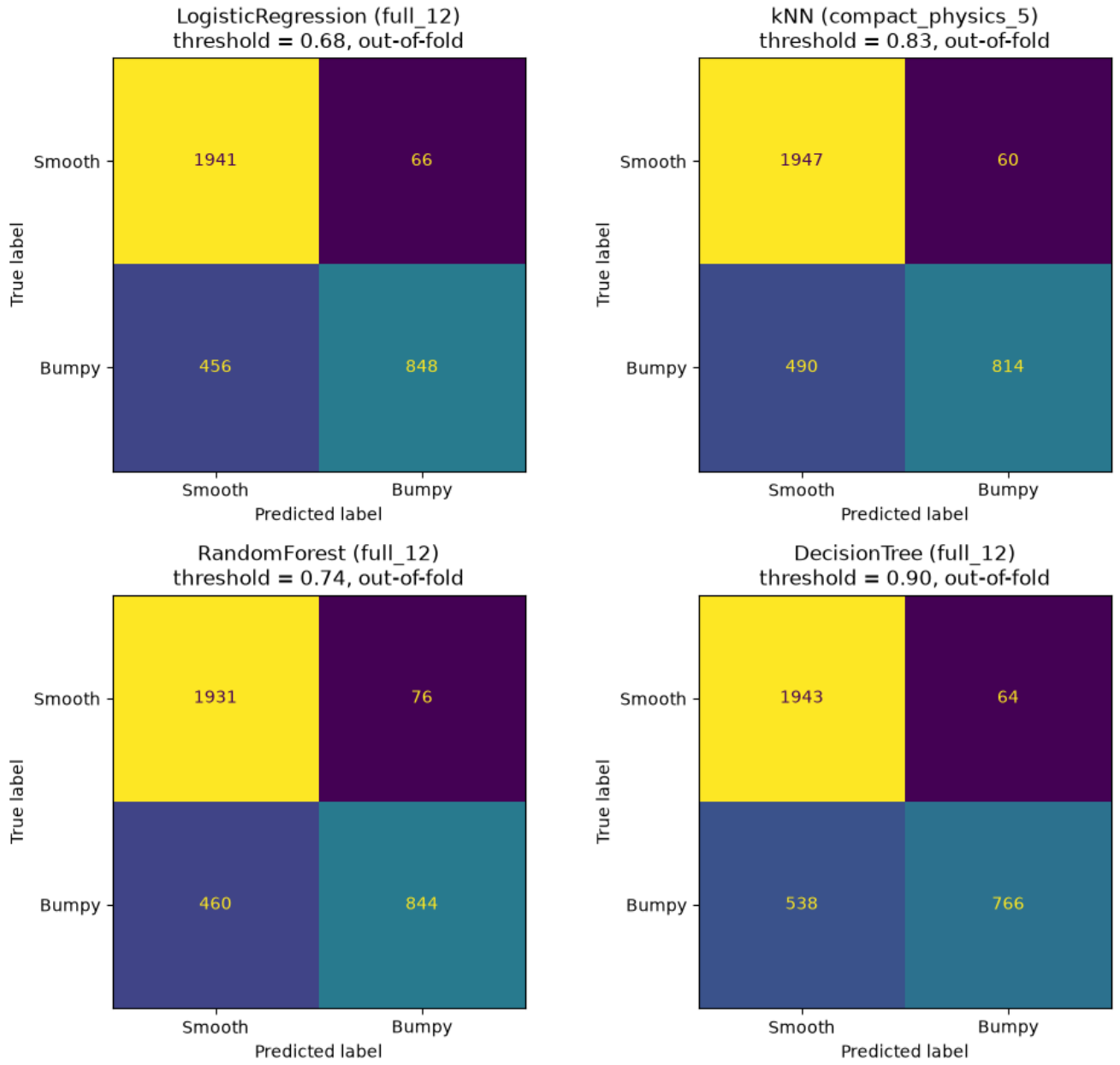


Figure 10: Training-only out-of-fold confusion matrices for the best configuration from each supervised model family. The thresholds shown were selected using out-of-fold training predictions and were fixed before final held-out evaluation. Logistic Regression achieved the highest validation $F_{0.5}$ score and was therefore selected as the final model.